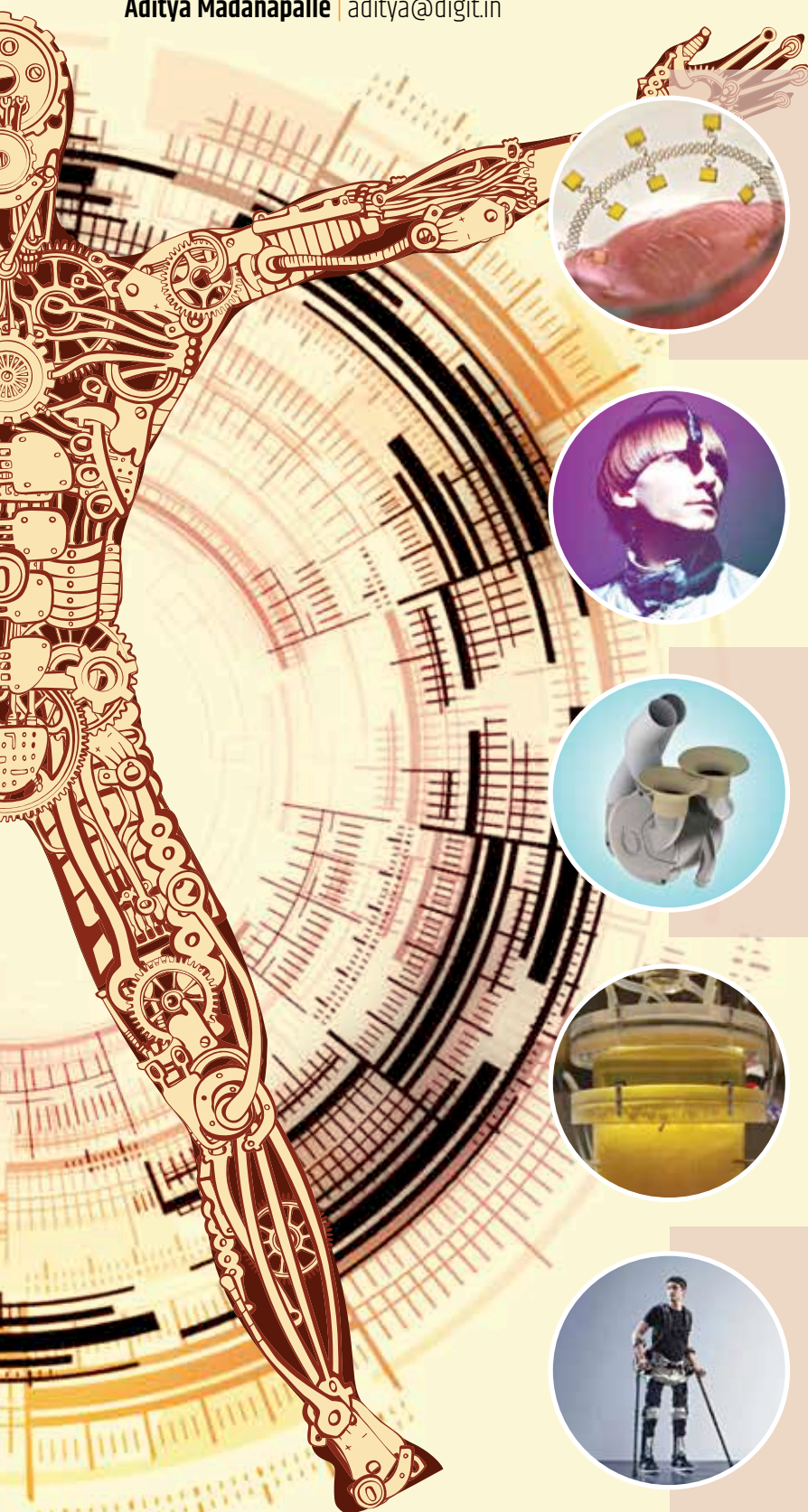


The human body can be thought of as a machine made up of flesh and bones instead of steel and actuators. Scientists are working out technological interventions that can give our sub-systems upgrades, which can potentially make us transhuman.

Aditya Madanapalle | aditya@digit.in



The Eyes

Forget Google Glass, or head mounted augmented reality devices, smart contact lenses are on the way. Samsung and Sony have already filed a patent for such a device in 2016. These devices could record images with the blink of an eye, or provide smart overlays to real life. The overlays could help in everything from giving presentations, to shopping, to driving in low visibility conditions, seeing through rain or fog.



The Ears

Hearing aids have been around since the 1600s, when the first ear trumpets began to be used. Now the hearing aids are sleek and nearly invisible. Cochlear implants bypass the damaged ear parts and directly electronically stimulates the hearing nerve. Neil Harbisson is a real life cyborg who used the hearing ability to overcome his color blindness, with a device that converts colours to sound.



The Heart

Artificial hearts that could replace the real thing are considered the holy grail of modern medical science. Research into creating an artificial heart has been going on for years, with the devices usually being tested on animals. There are multiple projects around the world trying to realise this. One of the design constraints is to have as few moving parts as possible, to reduce wear and tear, and chances of breaking down.



The Liver

Current bioartificial livers are extracorporeal devices, such as the artificial lungs. Technology has not advanced enough to allow for an internal artificial liver. The first artificial liver was developed by Dr Kenneth Matsumura. These devices are used to help patients while they wait for a donor, or support the liver, and perform some of its functions while it regenerates. The liver is not yet completely understood, and we don't even know all of its functions, to artificially replace it.



The Skeleton

The skeleton is deep within the human body, and changing this would be a complicated and excruciating process. However, what can be done in a relatively easy manner, is to use an exoskeleton. These exoskeletons would restore movement to the paralysed, and allow aged people to perform basic tasks. Powered exoskeletons can also be used by emergency response personnel, soldiers and healthcare workers.

